

REPORT 4.1 - shot places: build v31 (Opus 7ad929c) and Fable's review

2026-09-28. Spec: docs/SPEC_shot-places.md. Built by Opus at high from af729df; one run, 14:59 to 15:33 CDT, 31 min and 455k sub-agent tokens by the harness count (n = 1). Filed by Fable from Opus's hand-back: the harness refused Opus's own write of this file ("Subagents should return findings as text, not write report files"). Sections 1 to 7 are Opus's findings; Section 8 is Fable's review.

1. What changed

File	Change
js/round/track-analysis.js	proposeHoleShots rewritten to spec 4.2 and 4.3. Every stop carries place (toHoleYd, lie, ground, onTee, onGreen, afterGreen). places grouped own / other / off-map, each in time order. Preselection: the sequence with the most falls in distance to the hole, found exactly; dwell breaks ties. The last-stop rule, the green-range rule and the holedOut cup are removed. PLACE_MIN_DWELL_S = 15.
js/ui/screen-play.js	openShotConfirm rebuilt: fixed tee row, place rows, typed rows (BALL NOT HERE, ADD A SHOT), OTHER PLACES ON THIS HOLE. applyHoleEntry writes his tee mark or the map tee, the track places, the typed shots, and never a track cup. lastMark carries its own undo, has no 20 s timer and is shown inside the green sheet. SAVE HOLE's UNDO is that banner. The tee nudge is not raised on a course with a map.
js/round/round.js	addMapTee (source: 'map', method: 'map', or no mark). rebuildCourseLearning skips a method: 'map' tee.
css/base.css	Row buttons in the shots sheet 48 px tall, 15 px text ("BALL NOT HERE" wrapped to three lines at 360 px before).
tools/detection-scoring.html	A table comparing the v30 picker (copied verbatim from af729df) with the new preselection.
test/run.js, test/index.html	runShotPlacesTests: the ten spec tests. Two expectations updated, four tests retired (Section 2).
js/data/build.js, sw.js	v31 in both. REVISION untouched.

2. Tests

- At af729df: 571/571. The ten new tests against the unchanged modules: 571/581, the ten failing on their own behaviour. Final: **577/577**.
- Updated: the green flow's row 1 ends "(scorecard)"; its provenance reads map, track, gps, manual.
- Retired, because spec 4.3 and 5.1 replace the rules they assert: "the tee shot and the approach outrank sitting in the cart"; "the stop he holed out at is never offered as a full shot"; "the cup is taken from the retrieval, not from wherever the window ends"; "green stops come back rather than manufacture a shortfall".

3. The hole 12 trace (spec 6.1), done before any change

Round r_b440472a, recorder events.csv and fixes.csv, times CDT.

Time	What happened
11:59:42	Screen on.
12:00:09	Hole 12's tee recovered with USE THE TRACK (inferred: 'track').
12:00:30 to 12:00:50	Cart to the 13th tee.
12:01:27.9 to 12:01:30.9	Cup burst on hole 12, 4 samples.
12:01:31	Cup saved; setCup stamped completedAt; no putts entered.
12:01:51	Screen off.

Reproduced on the v30 modules: the recovered tee made hole 12 "started", so the next-hole arrow does not advance; promptGreenEntry opens "Hole 12 - putts", whose first control is MARK CUP, primary. Tapping it runs a cup burst on hole 12 from the 13th tee. CUP_AT_TEE_M stays silent (it measures from hole 12's recovered tee). The "Cup marked here. UNDO" banner sat under the sheet's scrim and the 20 s timer removed it. There is no tap log, so which tap he made is [inferred], n = 1.

In v31 the UNDO is in the green sheet, stays until the next mark, save or hole change, and restores the cup and completedAt. The prompt's MARK CUP button is unchanged: the spec does not touch it.

4. The spec's verify items

Item	Answer		
3.3 addShot with mark: null	Accepted; a mark is built only when reduced is given.		
3.4 readers of source / mark.method	lieUnanswered asks only 'gps' shots. holeHasTrackShots has no caller. The export writes rounds verbatim; import validates neither key. The strokes-gained input reads neither key and uses the position, as for a track shot. rebuildCourseLearning did learn any tee with a mark; it now skips method: 'map'.		
4.4 distanceEntry unit 'yards'	Accepted today (PUTT_UNITS.yards). A typed 140 yd is stored distanceFt: 420, { value: 140, unit: 'yards' }.		
6.4 a cup alone never completes a hole	Holds: isHoleComplete is 'greenEntry		manual. setCup stamps completedAt; the cup's UNDO restores it.

5. Scoring, tools/detection-scoring.html, the four labelled rounds

A pick counts when it lands within 10 m of his marked full shot. The old picker takes every full shot from the track; the new rule takes shot 1 from the map, so the like-for-like column is shots 2+. 2026-09-27 is not used.

Round	Map	Holes	Old: all full shots	Old: shots 2+	New: shots 2+	Map tee within 10 m of his tee mark (median)
FT4 Radcliffe	no	9	10/18	6/9	2/9	-
FT5 Veenker back 9	yes	9	15/26	10/17	9/17	5/9 (9.3 m)
FT6 Veenker 14-18	yes	5	16/18	12/13	10/13	2/5 (10.9 m)
FT6 Veenker hole 1	yes	1	2/3	1/2	1/2	1/1 (8 m)

At Veenker the new rule scores lower: 20/32 (63%) on shots 2+ against 23/32 (72%) for the old picker (n = 32 shots, 15 holes, 3 rounds).

Cause of the 12 misses	Misses
The shot's stop reads as on a green polygon ("not onGreen")	4
The stop starts after he first stood on the green ("not afterGreen")	3
The stop is on another hole's ground while this hole's pool was full	3
A stop just outside the tee polygon took a slot	2

Limits: these windows are bounded by his own marks, tighter than a pocketed hole's window. Radcliffe has no map, so the new rule has no signal there.

6. Where the spec was silent (Opus's choices)

1. No-map course: every stop is "other", the preselection falls to dwell ties, shot 1 is the card yardage with no position.
2. The tee nudge is not raised on a map course.
3. Map tee accuracy is the box's centre-to-corner reach (17.1 m, hole 1 gold); the shot list shows "from the map" instead of "poor fix".
4. Lists are in time order; toHoleYd orders only the preselection.
5. The falls start from the card yardage, or his tee mark's distance to the green centre.

6. A one-full-shot hole opens with just the tee row, even with no track.
7. NOT A SHOT is always offered; PLAYED TWICE beside it while short.
8. UNDO clears on a new mark, the green SAVE, the hand-entry SAVE, SAVE HOLE, the footer UNDO and a hole change.
9. A tee recovered from the track is not his mark; ENTER SCORE replaces it with the map tee.

7. Follow-ups noticed, not fixed

- The green-entry prompt still opens with MARK CUP as its first primary button: the hole 12 control.
- The lie row in the shots sheet overflows at 360 px ("Recovery" clipped); same call at v30.
- UNDO does not reverse learnCup / learnTee; the footer UNDO has the same gap.
- roundTotals.gpsShots counts map and track shots as GPS shots.
- Re-entering a hole saved before v31 keeps its old track cup.
- Edit mode has no map, so ENTER SCORE on a finished round saves shot 1 with no position.
- With no box for the tee set (hole 9, hole 16 blue) shot 1 has no position.

8. Fable's review (medium), 2026-09-28

Check	Result
Diff against the spec	track-analysis.js and round.js read line by line; screen-play.js read on code lines. Matches Sections 3 to 6. No departure found.
Must-not-touch list (spec Section 9)	9 files changed; none in android/, js/analysis/, js/gps/, js/util/geo.js, the data rails, the geometry, docs/roundDownloads/. REVISION untouched. BUILD.id and the cache both v31.
Suite	577/577 at 360x728, Fable's run, n = 1.
Commit identity	rusty9645@gmail.com.

Verdict: built to spec - PASS. The preselection rule in the spec - NOT GOOD ENOUGH, and that is the spec's fault, not the build's.

- On the labelled rounds it picks 3 fewer shots than the picker it replaces (20/32 against 23/32, n = 32).
- Replayed on the 2026-09-27 track (not labelled, illustration only): hole 14 offers a stop 377 yd from the green on a 397 yd hole as shot 2, and hole 1 offers one at 402 yd on a 419 yd hole. Both are stops beside the tee. That is the fault he named: *"the proposed tee shots on those holes were stupid"*, moved from shot 1 to shot 2.
- What v31 does deliver regardless of the preselection: the tee from the scorecard, every stop reachable, no cup from the track, BALL NOT HERE, and UNDO that stays.

Next: spec revision 4.2 (docs/SPEC_shot-places.md Section 11), three band-aware changes to the pool, built by Opus and scored the same way. v31 is not installed on his phone.